

NAME : _____

CLASS : _____



JURONG PIONEER JUNIOR COLLEGE
JC2 Preliminary Examination 2022

BIOLOGY
Higher 2

9744/02
13 September 2022

Paper 2 Structured Questions

2 hours

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your class and name in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use

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Total	

This document consists of **32** printed pages.

Answer **all** questions.

- 1 Fig. 1.1 is an electron micrograph of a cell in the pancreas with structures labelled **X** and **Y**.

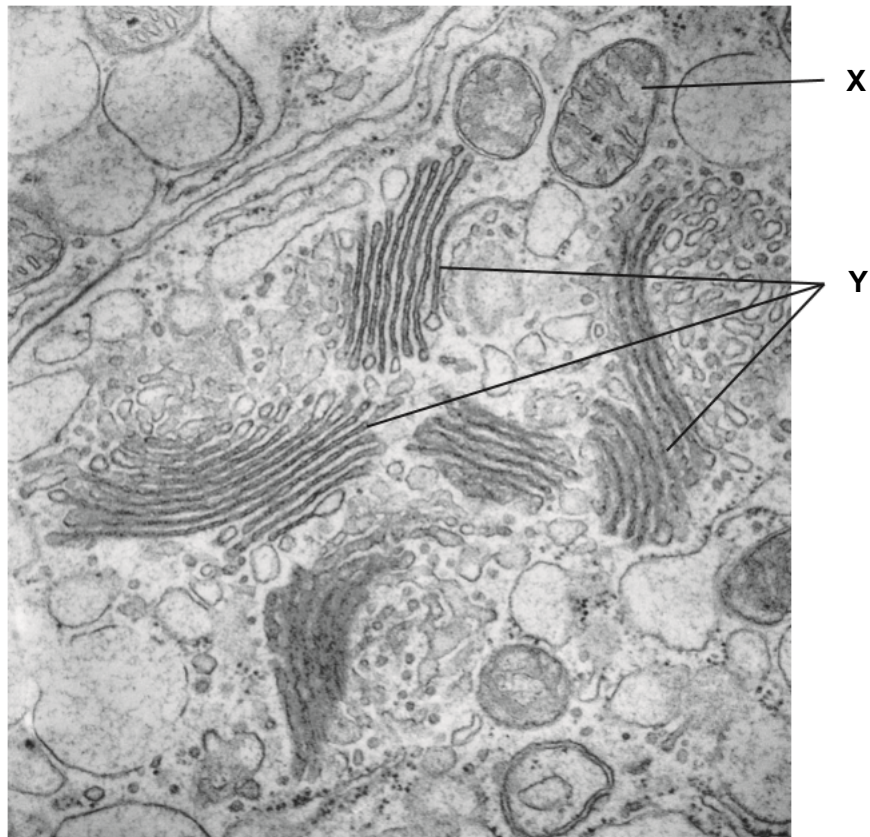


Fig. 1.1

- (a) Identify structures **X** and **Y**. In each case, outline one visible feature that allows the structure to perform its functions.

structure **X**

visible feature related to its functions

structure **Y**

visible feature related to its functions

.....[4]

1. Structure **X**: mitochondrion ;
2. Highly folded inner membrane / cristae for embedding electron carriers / ATP synthase for ATP synthesis ;
3. Structure **Y**: Golgi apparatus/body ;
4. Stack of flattened, membrane-bound sacs / cisternae for further modification / packaging of proteins ; OR
5. Swellings at end of sacs for vesicle formation /vesicles at ends of sacs ;

MP 4 or 5 max 1

Protein production involves a complex sequence of events and number of cell structures. Cells in the pancreas secrete enzymes, such as amylase, into a duct.

- (b) The first column in Table 1.1 shows some of the events that occur in the production of amylase in cells in the pancreas and its eventual release from the cell.

Table 1.1

event	sequence of events (numbers)	cell location (letters)
exocytosis		F
protein modification		
secretory vesicle formation		
transcription		
translation		

- (i) In Table 1.1, write the sequence in which the events occur, using 1 as the first process in the sequence. [1]

1. correct order of processes (5, 3, 4, 1, 2) ;

- (ii) From the list **A** to **F**, choose **one** cell location for each event and write the letter in Table 1.1. Each letter may be used once, more than once, or not at all. The first example, **F**, has been completed for you.

A Golgi apparatus

B lysosome

C nucleus

D rough endoplasmic reticulum

E smooth endoplasmic reticulum

F cell surface membrane

[1]

1. A and/or D, A, C, D ;

event	sequence of events	cell location (letters)
exocytosis	5	F
protein modification	3	A and/or D
secretory vesicle formation	4	A
transcription	1	C
translation	2	D

- (c) Explain how amylase that are secreted by cells in the pancreas are packaged into vesicles.

..... [4]

Answer can be given with reference to amylase/protein

1. Transport vesicle (containing proteins) budded off from cisternae/membrane of rER and moves proteins to the cisternae at the cis face of GA ;
2. (Membrane of) transport vesicles will fuse with (the membrane of) the cisternae at the cis face of the Golgi apparatus/body ;
3. Proteins are further modified, sorted and packaged into secretory vesicles ;
4. Secretory vesicles containing the modified/mature proteins bud off from the cisternae (at the trans face) of GA ;
5. Vesicles are directed to the cell surface membrane via microtubules ;
6. Energy / ATP is required (for movement of vesicles / fusion with membrane) ;
7. Amylase/protein is released via exocytosis ;

Mention "bud off" at least once in MP1 or MP4

[Total: 10]

2 Fig. 2.1 shows a section of the cell membrane of a red blood cell.

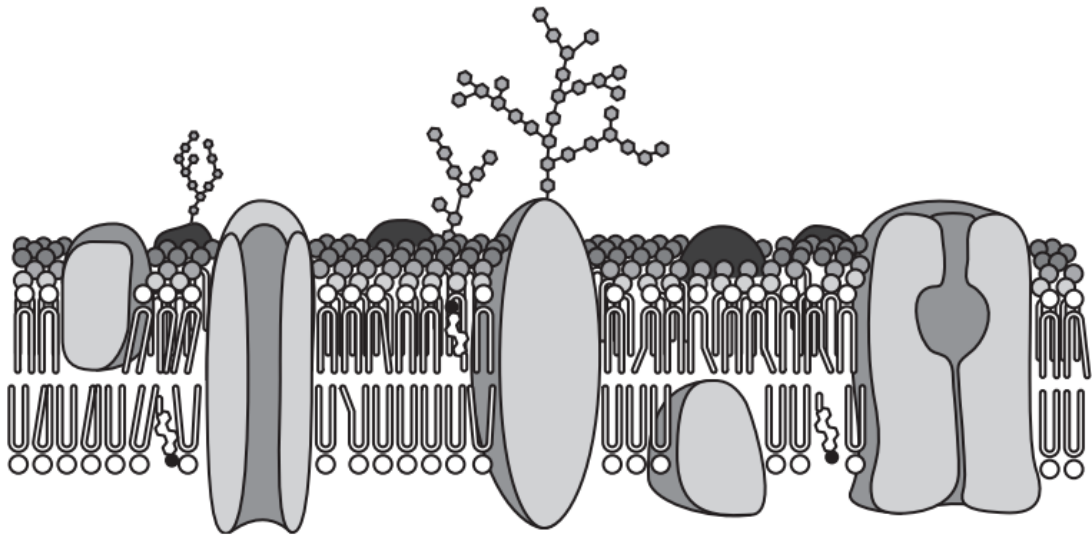


Fig. 2.1

- (a) The arrangement of the various constituent biomolecules supports the fluid mosaic model of the cell membrane.

Explain why the cell membrane is referred to as fluid mosaic.

.....[3]

Fluid

1. "Fluid" means that the phospholipids and proteins are free to move within the membrane ;
2. Phospholipids are held by (weak) hydrophobic interactions and move about rapidly by diffusion in their own layers ;

Mosaic

3. Proteins are embedded in the phospholipid bilayer in a random/scattered manner ;

Cyanide is a poison that inhibits cytochrome c oxidase involved in aerobic respiration.

Fig. 2.2 shows how cyanide concentration affects the uptake of chloride ions by red blood cells. The rates of chloride ion uptake are given as percentages of those obtained in a control experiment with no cyanide.

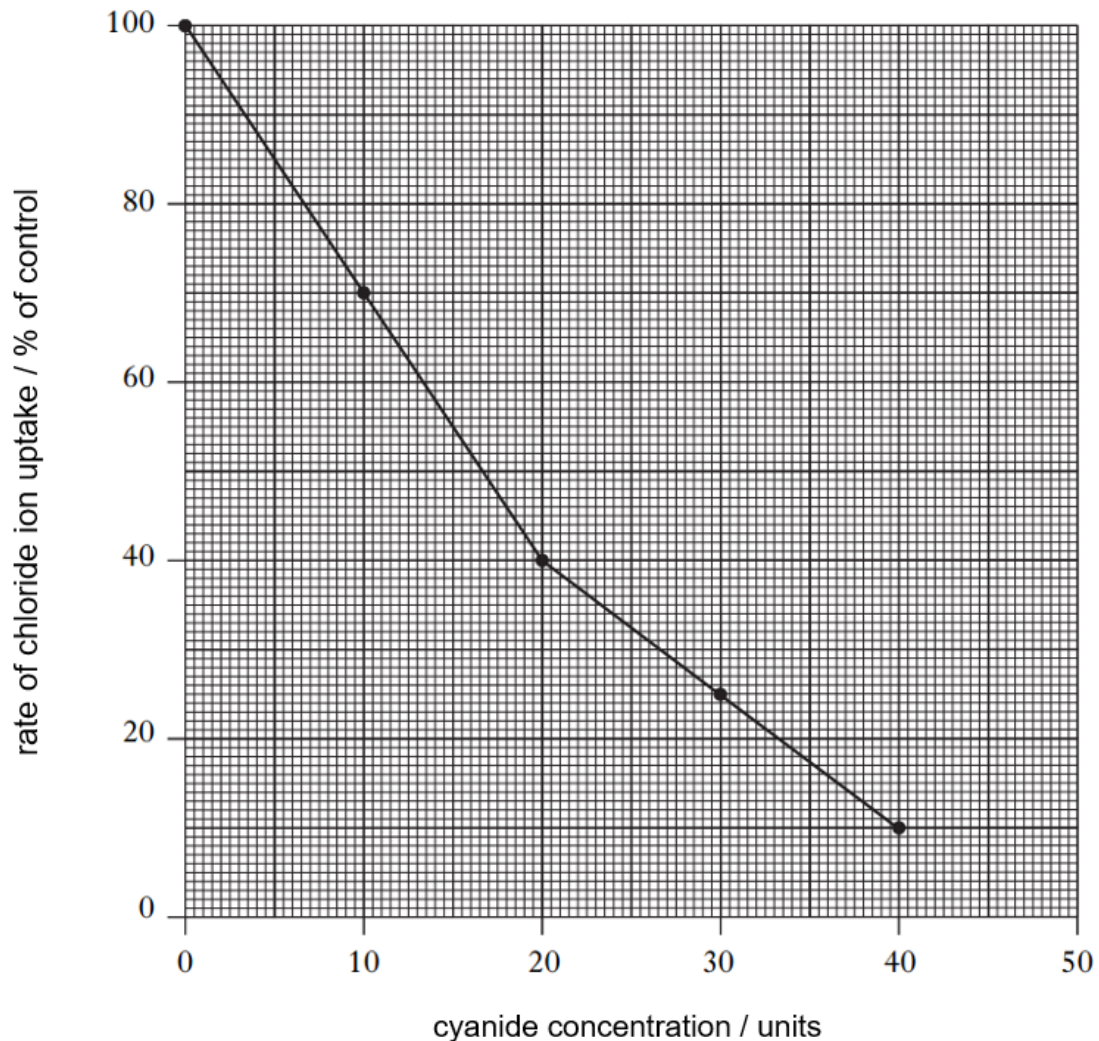


Fig. 2.2

- (b) It is known that the uptake of chloride ions requires the presence of transmembrane proteins.

Explain why chloride ions cannot freely pass through the phospholipid bilayer of membranes.

.....[2]

1. Non-polar/hydrophobic fatty acids make up the hydrophobic core of the cell membrane/phospholipid bilayer of membranes, thus the membrane is impermeable to ions ;
2. Cl^- are charged and cannot move across the membrane because they are repelled by the hydrophobic core of the phospholipid bilayer of membranes ;

(c) With reference to Fig. 2.2,

(i) describe how cyanide concentration affects the rate of chloride ion uptake

..... [2]

1. As cyanide concentration increases from 0 units to 20 units, rate of chloride ion uptake decreases sharply from 100% of control to 40% of control ;
2. As cyanide concentration increases from 20 units to 40 units, rate of chloride ion uptake decreases gradually from 40% of control to 10% of control ;

(i) identify how chloride ions are moved across the membrane of the red blood cells

..... [1]

1. active transport ;

(ii) account for your answer in (c)(i).

..... [2]

1. When cyanide concentration increases, rate of aerobic respiration decrease leading to decrease in (amount of) ATP synthesised ;
2. Decrease in ATP synthesis is correlated to decrease in rate of chloride ion uptake, indicating that ATP is required for the transport of Cl⁻ (as active transport involves ATP expenditure) ;

[Total: 10]

- 3 The DNA double helix can be successfully compacted into a nucleus with a diameter that is only a millionth of the length of DNA.

(a) Explain how DNA can be successfully compacted into a nucleus.

..... [4]

1. **Negatively charged DNA molecule winds around a positively charged histone octamer consisting of 2 molecules of H2A, H2B, H3 and H4 each to form a nucleosome ;**
2. **Individual nucleosomes are connected by strands of linker DNA and H1 histones to give (10nm) 'beads-on-a-string' / nucleohistone complex ;**
3. **With the aid of H1 histones, the 'beads-on-a-string' / nucleohistone complex coils (R: fold) to form a (30nm) chromatin fibre ;**
4. **The 30nm chromatin fibre folds (R: coil) to form looped domains that are attached to a base of scaffolding proteins / non-histones, forming the (300nm) chromatin fibre ;**
5. **that is further coiled and compacted to form the (highly-condensed) chromosome ;**

MP 1-5: max 4

Semi-conservative DNA replication occurs in the nucleus. This begins at the origins of replication, where the enzyme helicase causes the DNA molecule to unwind and unzip, causing the two parental strands to separate.

(b) Outline the role of **three other** enzymes in DNA replication.

..... [3]

1. **Primase catalyses the formation of a short RNA primer ;**
2. **DNA polymerase then binds to the RNA primer and adds free DNA nucleotides to the free 3' –OH end of the RNA primer / synthesises the new DNA strands in the 5' to 3' direction ;**
3. **DNA polymerase catalyses the formation of phosphodiester bonds between adjacent DNA nucleotides ;**
4. **DNA polymerase synthesises the leading strand continuously and the lagging strand discontinuously resulting in Okazaki fragments ;**
5. **Another DNA polymerase replaces RNA nucleotides of the RNA primers with DNA nucleotides ;**
6. **DNA ligase seals the gaps between DNA fragments by catalysing the formation of phosphodiester bonds between adjacent DNA nucleotides to form a continuous strand ;**

MP 2-4: max 1

Mention catalyse at least once for full credit

Besides DNA replication, transcription also occurs in the nucleus.

- (c) Complete Table 3.1 using a tick (✓) to indicate which features apply to each of the processes. Use a cross (✗) for features that do **not** apply. The first row has been completed for you.

Table 3.1

feature	DNA replication	transcription
a single-stranded molecule is produced	✗	✓
hydrogen bonds are broken		
both strands of DNA act as templates		
covalent bonds are formed		

[3]

feature	DNA replication	transcription
a single-stranded molecule is produced	✗	✓
hydrogen bonds are broken	✓	✓
both strands of DNA act as templates	✓	✗
covalent bonds are formed	✓	✓

1 mark for each correct row ;

[Total: 10]

- 4 Fig. 4.1 shows some of the main structural features of an influenza virus. PA, PB1 and PB2 are RNA polymerases.

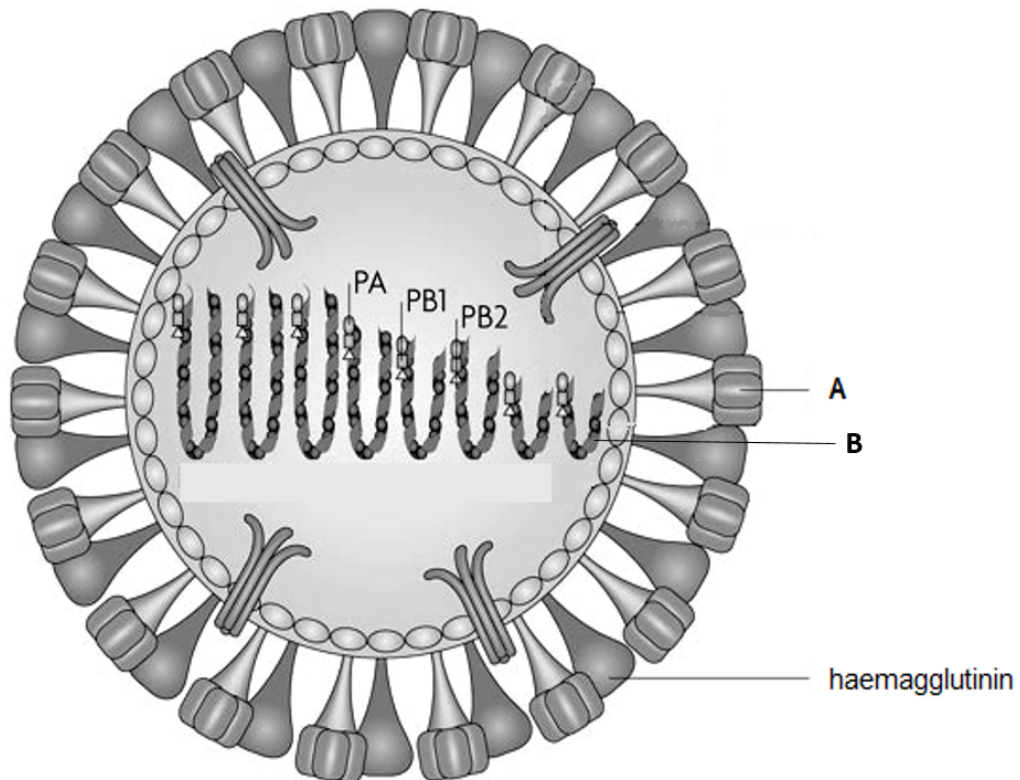


Fig. 4.1

- (a) Identify the labelled structures and state their functions.

structure **A**

function

.....

structure **B**

function

.....

[4]

1. **A – neuraminidase ;**
2. **It (is an enzyme that) catalyses the cleavage of sialic acid residues from haemagglutinin ;**
OR
3. **facilitating the exit and release of newly replicated influenza viruses from the (infected) host cell by budding ;**
4. **B – (single stranded) RNA (genome) ; *l: double stranded***

5. Viral RNA serves as the template for the synthesis of complementary RNA (cRNA), (using the viral RNA-dependent RNA polymerase) to express viral gene products (RNA and proteins) ;

(b) Explain why the influenza virus requires its own RNA polymerases PA, PB1 and PB2.

.....[1]

1. These RNA-dependent RNA polymerases are not present in the host cell
A: host cell's RNA polymerases are DNA-dependent RNA polymerases ;

(c) Discuss how viruses challenge the cell theory.

..... [2]

1. Cells are basic units in living things, whether unicellular or multicellular but viruses are acellular (no cytoplasm or organelle) ;
2. Cells contain hereditary information in DNA which is passed from cell to cell during cell division but viruses contain only one type of nucleic acids/either DNA or RNA ;
3. All cells arise from pre-existing cells by division but viruses must replicate using the host cell's metabolic machinery/enzymes and ribosomes ;

- (d) Oseltamivir is an antiviral inhibitor used for the treatment of infection with influenza viruses.

Fig. 4.2 shows the number of copies of influenza genome per cm^3 plasma in two groups of infected patients – one group treated with oseltamivir and another control group without oseltamivir.

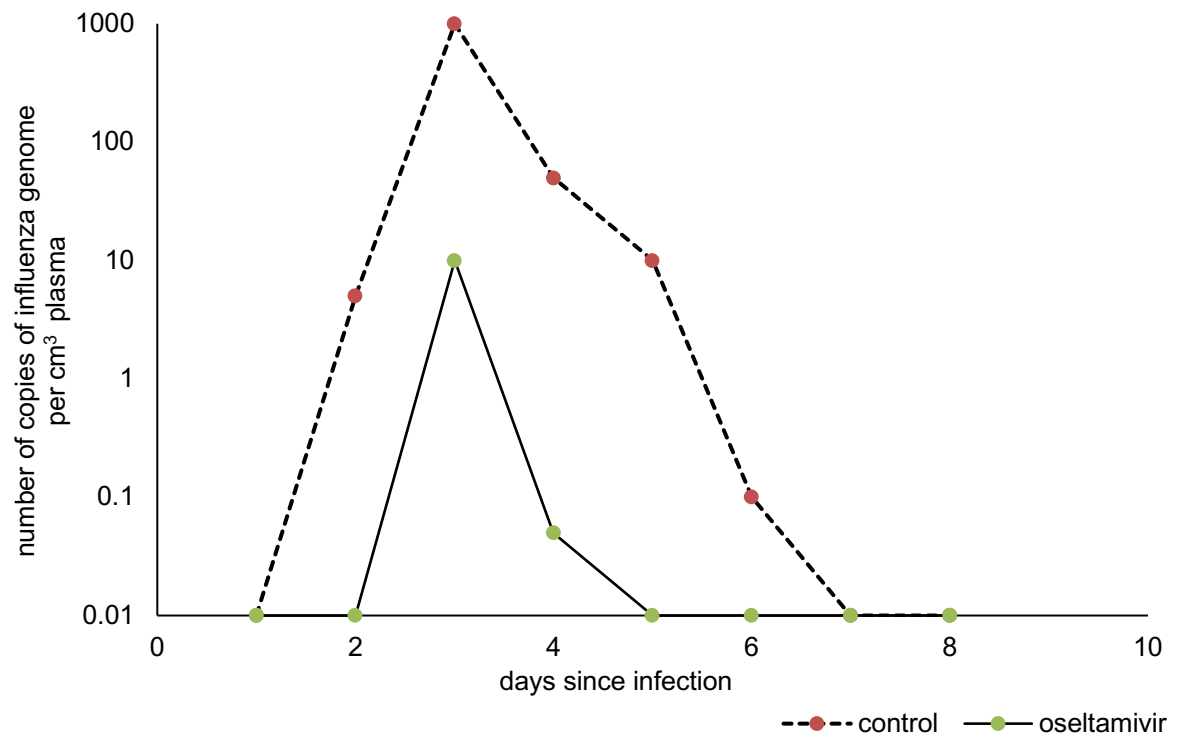


Fig. 4.2

Describe the effect of oseltamivir on the number of copies of influenza genome in patients, as shown in Fig. 4.2.

..... [3]

1. Maximum number of copies of influenza genome per cm^3 plasma reduced from 1000 to 10, 3 days since infection ;
2. Number of copies of influenza genome per cm^3 plasma lowered to 0.01 by day 5 as compared to day 7 of control/ not treated with oseltamivir ;
3. Influenza replicates for 3 days for those treated with oseltamivir vs. 6 days of those in control ;

[Total: 10]

- 5 (a) Describe **two** ways in which the *lac* operon is **similar** to the *trp* operon.

..... [2]

1. Both operons contain a cluster of structural genes that make up a single transcription unit ;
OR
2. Both operons contain a cluster of structural genes that are transcribed as a single / polycistronic mRNA ;
3. (Expression of) Structural genes is under the control of the same promoter and operator ;
4. The structural genes code for enzymes / functionally related proteins involved in the same / a single metabolic pathway ;
5. When active repressor binds to the operator, it inhibits the transcription of the structural genes / operon is switched off ;

MP1 and MP2: max 1

- (b) Describe **four** ways in which the *lac* operon is **different** from the *trp* operon.

..... [4]

contrasting feature	<i>lac</i> operon	<i>trp</i> operon
1. type of operon ;	inducible	repressible
2. default state of operon expression ;	off (in the absence of inducer)	on (in the absence of co-repressor)
3. effector molecule ;	<u>allolactose</u> acts as an <u>inducer</u>	<u>tryptophan</u> acts as a <u>co-repressor</u>
4. effect of effector molecule on operon ;	(in the presence of inducer, allolactose) operon / transcription of structural genes is turned <u>ON</u>	(in the presence of co-repressor, tryptophan) operon / transcription of structural genes is turned <u>OFF</u>
5. default conformation of repressor ;	repressor synthesised in the active conformation	repressor synthesised in the inactive conformation
6. metabolic pathway involved ;	(protein products of operon involved in) catabolic pathway / breakdown of lactose	(protein products of operon involved in) anabolic pathway / synthesis of tryptophan

(c) Suggest the advantages to bacteria of arranging some genes in operons.

..... [2]

1. Operons allow for the simultaneous regulation of related genes with related functions / which are involved in the same metabolic activity ;
2. Genes coding for proteins/enzymes of a single biochemical pathway / of related functions / same metabolic activity (such as the catabolism of carbohydrates) are grouped together into an operon for easier control ;
3. Operons enable the simultaneous regulation of related genes to adapt and respond to environmental changes ;
4. The genes on the operons are only expressed when required, allowing the bacteria to produce the proteins/enzymes that are required for increasing efficiency / preventing wastage of energy and resources.
5. Bacteria are able to utilise a variety of carbohydrates as respiratory substrates (for their energy requirement) ;
6. This confers selective advantage to the bacteria ;

MP6 will only be awarded provided there is some prior elaboration and logical link provided (i.e. only if reference to any of the other MPs was made)

[Total: 8]

- 6 The polymerase chain reaction (PCR) is commonly used in medical and biological research to produce large quantities of DNA from a very small original sample.

(a) The main steps of one PCR method are shown in Fig. 6.1.

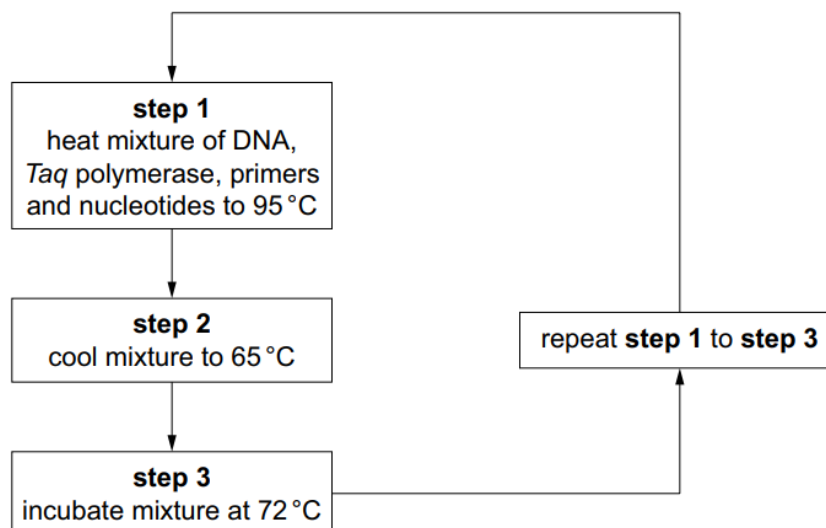


Fig. 6.1

- (i) Explain why it is necessary to heat the mixture to 95 °C (**step 1**).

.....[2]

1. To denature DNA / to separate the two DNA strands / double stranded DNA ;
2. By breaking hydrogen bonds between complementary base pairs (of the double-stranded DNA) ;
3. So that the bases are exposed to produce single-stranded DNA / template strands (for complementary base pairing with dNTPs) ;

MP 1-3: max 2

- (ii) Explain why the enzyme *Taq* polymerase, rather than any other type of DNA polymerase, is used in PCR.

.....[2]

1. *Taq* polymerase is heat stable / works at high temperature ;
2. So it does not need to be replaced during each cycle of PCR ;
- OR
3. Other DNA polymerases are not heat stable ;
4. So they would have to be replaced during each cycle of PCR ;

Another molecular technique used in medical and biological research is Southern blotting followed by nucleic acid hybridisation.

To visualise the results, autoradiography is carried out at the end. The images formed on the photographic X-ray film would correspond to the bands that contain the DNA sequence of interest.

(b) Outline the process of Southern blotting and nucleic acid hybridisation.

..... [4]

1. **Gel** from gel electrophoresis is placed on top of a sponge soaked in alkaline solution, a **nitrocellulose membrane**, followed by a stack of **paper towels**, are placed **on top of the gel** ;
2. **Alkaline solution** is drawn upwards through the gel, **denaturing** the (double stranded) **DNA** fragments, **transferring single-stranded fragments of DNA to the nitrocellulose membrane** (where they adhere firmly in the same positions as they were in the gel) ;
3. Nitrocellulose membrane is placed in a sealed plastic bag containing **radioactive single-stranded DNA probes** (complementary to the DNA sequence of interest) ;
4. Which hybridise with DNA fragments (containing sequence of interest) on the nitrocellulose paper via **complementary base pairing** ;
5. Nitrocellulose membrane is removed from the bag and washed thoroughly to **remove any unhybridised probes** ;
Autoradiography is then carried out to visualise the results.

MP 1-5: max 4

Sickle cell anaemia is a genetic disease caused by a recessive mutant allele of the β -globin gene.

Scientists investigated the inheritance of the disease in one family. DNA was collected from each family member. PCR, restriction enzyme digestion using the enzyme *DdeI*, gel electrophoresis, Southern blotting and nucleic acid hybridisation were carried out.

Fig. 6.2 shows the autoradiography results from the final stage of this investigation.

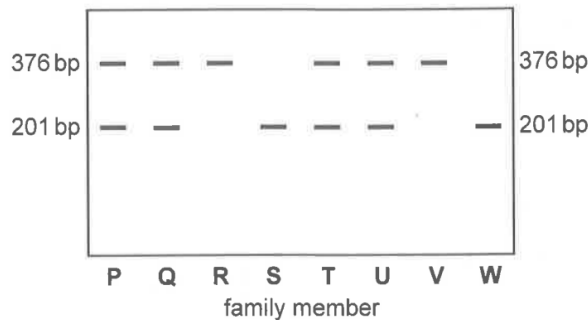


Fig. 6.2

Fig. 6.3 shows where the restriction enzyme *DdeI* cuts within the two different β -globin alleles and the sizes of the fragments produced.

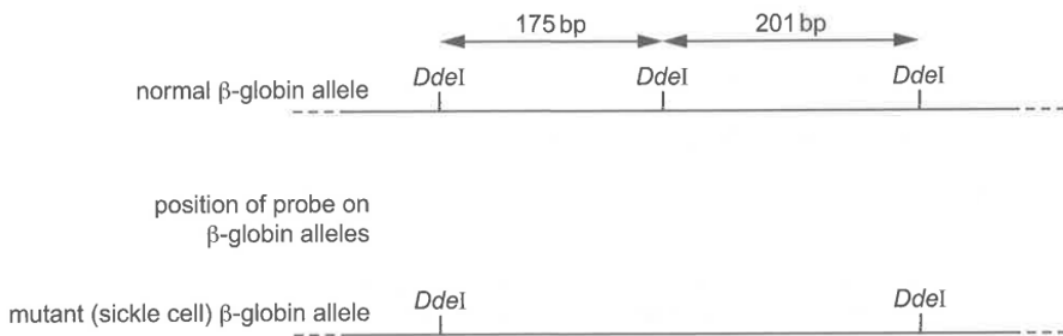


Fig. 6.3

(c) Using the information from Fig. 6.2,

(i) indicate in Fig. 6.3 the position of the probe that was used [1]

1. Indicates a region corresponding to the 201 bp fragment AND that does not overlap with any region in the 175 bp fragment of the normal β -globin allele ;

(ii) state which family members suffer from sickle cell anaemia.

.....[1]

1. Individual R and individual V ;

[Total: 10]

7 Fig. 7.1 shows an animal cell in a stage of the mitotic cell cycle.

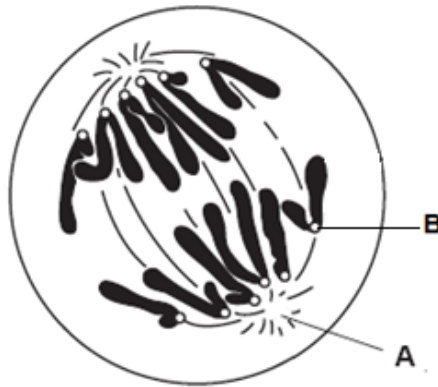


Fig. 7.1

(a) A pair of rod-like structures can be found in region **A**.

State **one other** feature of these structures and outline their role during mitosis in animal cells.

feature

role

[2]

Feature

1. (The centriole pair is) positioned at right angles / perpendicular to each other ;
2. (transverse section shows) 9 triplets of microtubules arranged in a ring ;

Role

3. The two pairs of centrioles move to the opposite poles of the cell (during prophase) and determine the polarity of the cell ;
4. Centrioles act as the microtubule-organising centre (MTOC) – the centrioles produce spindle fibres (radiating from the centrioles) at the poles towards the equator of the cell ;
5. Centrioles organise the synthesis of spindle fibres which lead to the separation of (sister) chromatids during mitosis ;

MP1/MP2 + any from MP3-5 for full credit

(b) Identify structure **B** and outline its structure and function.

B

structure

function

[3]

B

1. **Centromere ;**

Structure

2. **A constricted region on a chromosome ;**
3. **Consists of non-coding DNA made up of a series of tandem repeats / is (often) heavily methylated ;**
4. **Site at which (a number of centromere-associated) proteins bind, forming (a specialised structure called) the kinetochore ;**

Function

5. **Centromeres hold sister chromatids together (until anaphase of mitosis) ;**
6. **Centromeres divide to allow separation of sister chromatids (during anaphase of mitosis) ;**
7. **DNA at the centromere is essential for proper alignment and separation of sister chromatids to opposite poles of the cell during nuclear division ;**
8. **Centromeres ensure proper segregation of chromosomes by being the site on the chromosome where the kinetochore assembles and microtubules/spindle fibres from the centrioles attach during nuclear division ;**

MP1 + any from MP2-4 + any from MP5-8 for full credit

- (c) With reference to Fig. 7.1, state **two** observable differences between the behaviour of chromosomes in mitosis and meiosis.

..... [2]

	mitosis	meiosis	
1.	(during anaphase) <u>sister chromatids</u> separate	(during anaphase I) <u>homologous chromosomes</u> separate	;
2.	(during anaphase) division of the centromere occurs	(during anaphase I) no division of centromere	;

- (d) The stages and checkpoints of the mitotic cell cycle are closely regulated. Explain the need to regulate the mitotic cell cycle tightly.

..... [3]

1. The mitotic cell cycle is controlled by checkpoints such as the G₁ checkpoint / G₂ checkpoint / M checkpoint (at least 1 e.g.) which determine whether or not the cell cycle can proceed ;
2. *(mark for idea)* Requirements of each checkpoint must be met before cells move on to next stage, e.g. accurate completion of DNA replication / bring about DNA repair mechanisms if required / cell cycle arrested when there is DNA damage etc. ;
(A: tight control is needed for normal cell division and cell development with some elaboration)
3. Tight control of the mitotic cell cycle can thus prevent dysregulation of checkpoints of mitosis / cell division ;
4. preventing excessive cell cycle progression / rapid cell cycle / uncontrolled cell division which can lead to tumour formation and possibly cancer ;

[Total: 10]

- 8 The fruit fly, *Drosophila melanogaster*, feeds on sugars found in damaged fruits.

A fruit fly with normal features is described as wild type. It has red eyes and its wings are longer than its abdomen. There are mutant variations such as purple eyes or short (vestigial) wings.

Fig. 8.1 shows a wild type fruit fly and a mutant fruit fly with purple eyes and vestigial wings.

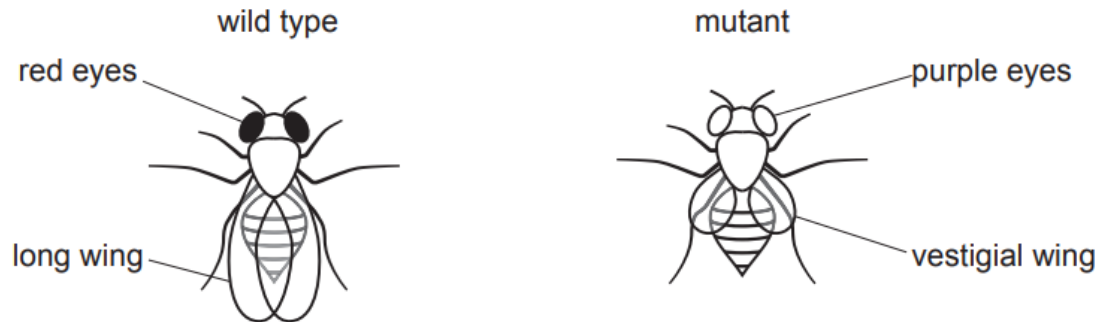


Fig. 8.1

- The genes coding for eye colour and wing length are located on the same chromosome.
- Allele **R** for red eyes is dominant to allele **r** for purple eyes.
- Allele **N** for long wings is dominant to allele **n** for vestigial wings.

- (a) A wild type fruit fly, heterozygous for both genes, was crossed with a fruit fly that was homozygous recessive for both genes.

Table 8.1 is a summary of the cross.

Complete Table 8.1.

Table 8.1

	wild type parent		double homozygous recessive parent	
parental phenotype	red eyes, long wings x purple eyes, vestigial wings			
parental genotype	$\frac{RN}{rn}$		x	$\frac{rn}{rn}$
offspring				
offspring genotype	$\frac{RN}{rn}$	$\frac{rn}{rn}$	$\frac{Rn}{rn}$	$\frac{rN}{rn}$
offspring phenotype	red eyes, long wings	purple eyes, vestigial wings	red eyes, vestigial wings	purple eyes, long wings
number of offspring	1339	1195	151	154

1. Correct parental phenotypes ;
2. Correct parental genotypes ;
3. Correct offspring genotypes (for all 4) ;
penalise under MP2 & MP3 if linkage notation not used
4. Correct offspring phenotypes (for all 4) ;

	wild type parent		double homozygous recessive parent	
parental phenotype	 X 			
parental genotype	 X 			
offspring				
offspring genotype				
offspring phenotype				
number of offspring	1339	1195	151	154

[4]

- (b) A statistical test showed that the results of the cross were significantly different from those expected.

State the name of the statistical test used and explain why it would be useful to carry out the test on these results. No calculations are required to answer this question.

statistical test

explanation

[2]

1. Chi-squared / χ^2 (test) ;
2. To estimate the probability that differences between observed and expected results were due to chance ;

- (c) Describe and explain the difference between the results of the cross in Table 8.1 and those expected.

..... [4]

Describe

1. Expected ratio from the cross is 1:1:1:1 but observed ratio was 9:8:1:1 ;
2. The observed offspring of the cross are majority of parental combinations/phenotypes (A: red eyes, long wing and purple eyes, vestigial wing) / minority of recombinant combination/phenotypes ;

Explain

3. The 2 genes for eye colour and wing length are linked / the alleles of the 2 genes for eye colour and wing length are inherited together as one linkage group ;
reference to both traits required for MP3
4. Resulting in a higher proportion of gametes/higher chances of getting gametes carrying the parental types ;
5. Crossing over between the 2 linked genes on non-sister chromatids of homologous chromosomes may occur ;
6. As crossing over is a chance event, this results in a lower proportion of gametes / lower chances of getting recombinant gametes ;
7. AVP – no independent assortment of homologous chromosomes occurs ;

At least 1 from MP1-MP2 required for full credit

[Total: 10]

- 9 The red poppy, *Papaver rhoeas*, and several species of daisy often co-exist as weeds of wheat fields.

Fig. 9.1 shows changes in the percentage frequency of red poppies and daisies in an area of wheat fields over a six year period from 1998 to 2003. From 1985, the herbicide metsulfuron-methyl was used to control weeds in this area of wheat fields. This practice continued throughout the six year period.

1998 showed the first occurrence of a red poppy known as biotype X. This red poppy had a specific mutation not present in normal red poppies.

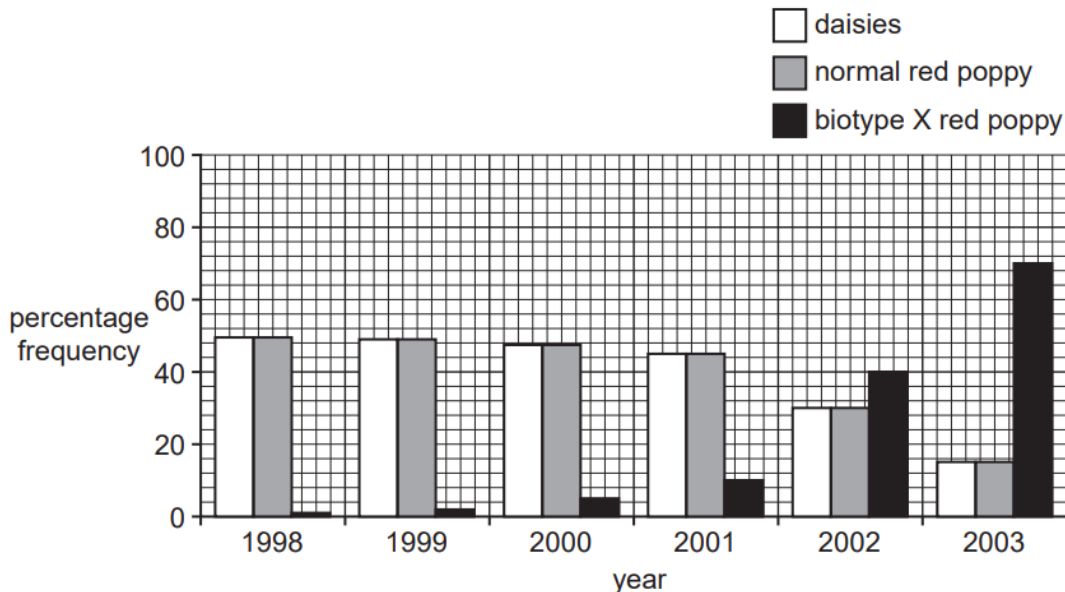


Fig. 9.1

- (a) Describe how the percentage frequencies of red poppies and daisies changed over the six year period.

..... [3]

1. From 1998 to 2003, percentage frequencies of daisies and normal red poppies decreased from 50% to 15% (A: 14-16%) ;
2. From 1998 to 2003, percentage frequencies of biotype X red poppy increased from 1% (A: 2%) to 70% ;
3. After 2001, there was a steep increase (30% increase by 2002 or 60% increase by 2003) in percentage frequencies of biotype X red poppy ;
4. From 1998 to 2003, total percentage frequencies of red poppies increased, from 51% (A: 52%) to 85% (A: 84-86%) ;
5. Percentage frequencies of daisies and normal red poppies were always equal throughout the six years ;

- (b) Suggest a reason for the distinct difference in percentage frequencies of normal red poppies and biotype X red poppy in the year 2003.

..... [1]

1. **Normal red poppies killed by / susceptible to metsulfuron-methyl but biotype X red poppy resistant to metsulfuron-methyl ;**
OR
2. **The specific mutation present in biotype X red poppy enabled it to survive well even in the presence of metsulfuron-methyl ;**

Although biotype X red poppy and normal red poppies continue to grow within the same geographical location, after numerous generations, biotype X red poppy may eventually become a different species from the normal red poppies.

- (c) Suggest why biotype X red poppy may eventually become a different species from the normal red poppies.

..... [2]

1. **Physiological / reproductive isolation (e.g. due to polyploidy, different flowering times) leading to sympatric speciation ;**
2. **Prevents interbreeding between biotype X and normal red poppies, leading to disruption in gene flow ;**

Daisies belong to the *Compositae* family, which is the largest family of flowering plants comprising more than 25 000 species. Many important evolutionary questions about the diversity in this family remain unanswered due to the lack of evidence to support major nodes of the phylogeny.

- (d) Explain the advantages of using molecular methods to determine phylogenetic relationships within the *Compositae* family.

..... [3]

1. **Unambiguous and objective. A, T, G, C are easily recognised and one cannot be confused with another / not dependent on subjective judgements or observations involving qualitative differences in morphology ;**
2. **Quantifiable and can be converted to numerical form and open to statistical analysis ;**
3. **Homologous regions of DNA from different species provides many points of comparison as each nucleotide position is a point of comparison ;**
4. **Molecular evidence avoids pitfalls of convergent evolution when morphological evidence could be due to convergence ;**
5. **Molecular / nucleotide data can be used to trace evolutionary relationships of species that are so different that there is little morphological homology ;**
6. **Molecular / nucleotide can be used to assess phylogenetic relationships that cannot be measured by comparative anatomy / other non-molecular methods e.g. fossils ;**

Fossil evidence indicates that flowering plants first appeared about 125 million years ago and they have since been rapidly diversifying. The comparison of chloroplast genomes could provide insight into the evolution of flowering plants.

- (e) Explain how evidence based on homologies identified in chloroplast genomes support Darwin's theory of evolution.

..... [2]

1. (Molecular) homology identified in the chloroplast genomes suggest common ancestry / descent from a common ancestor ;
2. Cumulative changes in chloroplast genome sequences show descent with modification ;
- OR
3. Comparisons of homologies in chloroplast genome between species show how a population may have been modified into descendent species (through natural selection and changes in allele frequency) ;

Although plant fossils are usually rare compared to fossils of bones, teeth and shells, there are many great fossil sites in the world that have excellent preservation of plant materials.

- (f) Suggest why plant fossils are usually rare.

..... [1]

1. Plant material (e.g. leaves) are soft and decompose rapidly ;

[Total: 12]

- 10** Measles is a common viral infection. A vaccine has been available for measles since the 1960s. There are vaccination programmes for many diseases including measles. Babies are born with a passive immunity to measles so the vaccine is not given in the first few months after birth.

(a) Explain how active immunity differs from passive immunity.

..... [3]

Feature	Active immunity	Passive immunity
Involvement of T or B cell ;	immune response involves clonal selection and expansion (A: activation) of B cells or T cells	does not involve B cells or T cells
Source of antibodies ;	antibodies are synthesised from plasma cells	antibodies are acquired from mother
Durability of immune response	long-lived effect of active immunity	short-lived effect of passive immunity
Lag time for immune response ;	immune response of active immunity is not immediate	immune response of active immunity is almost immediate
Memory cell production ;	memory cells are produced	memory cells are not produced
Removal of antibodies from circulation ;	antibodies are not removed	antibodies are removed

The World Health Organisation (WHO) publishes data on the vaccination programmes for infectious diseases. The WHO recommends vaccination rates of over 90% of children.

Each health authority in a country reports its success in vaccinating children in their district. The WHO uses these figures to estimate the percentage of districts in each country that vaccinate 90% of children against measles.

The WHO also collects statistics on death rates of children under the age of 5 from all causes, including infectious diseases.

Fig. 10.1 shows these statistics for 24 countries for the year 2007.

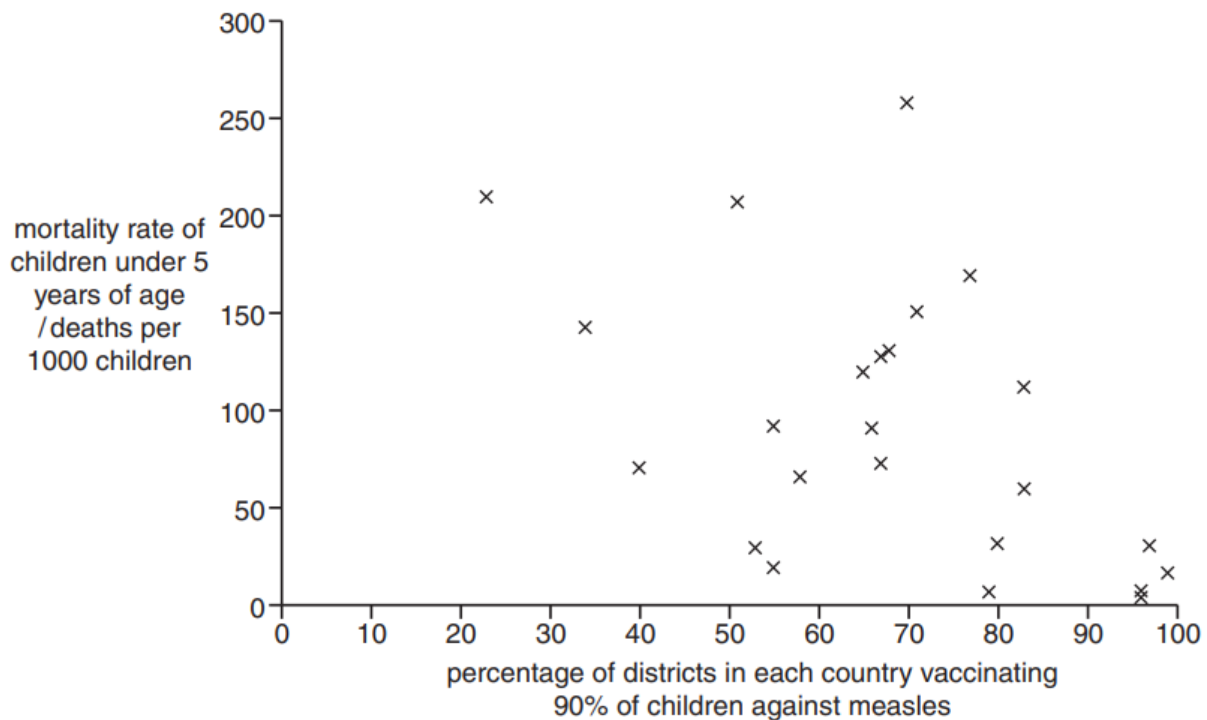


Fig. 10.1

- (b) Use the information in Fig. 10.1 to explain why the WHO recommends immunisation of 90% of children.

..... [2]

1. All countries with >90% (A: 96-99%) of districts reporting 90% of children vaccinated have very low death rates /5 deaths per 1000 children under 5 years of age ;
OR
2. Countries with less than 90% of districts (A: appropriate data e.g. 20% district), has high death per 1000 children (A: appropriate data e.g. 210 deaths per 1000 children under 5 years of age) ;
3. Herd immunity is achieved for countries with >90% district vaccination ;

MP 1 or 2 + MP 3 for max 2

[Total: 5]

11 The concept of climate change and global warming has been of concern to scientists for many years.

- (a) One way to collect data about atmospheric concentrations of greenhouse gases in the past is to study samples of ice from ice sheets in Antarctica. Ice samples from deep in the ice sheets were formed hundreds of thousands of years ago, while those near the surface were formed recently.

As ice forms, small bubbles of air are trapped in the ice. These air bubbles can be analysed to determine the concentration of carbon dioxide present. It is also possible to use chemical techniques to determine when the air bubbles were trapped.

Scientists studying climate change measured carbon dioxide concentrations in air bubbles from ice samples of known age, collected from near the surface.

Fig. 11.1 shows the concentration of carbon dioxide measured in air bubbles that were trapped in ice from 1959 to 1990. Direct measurements of atmospheric carbon dioxide from 1948 to 1978 are also shown.

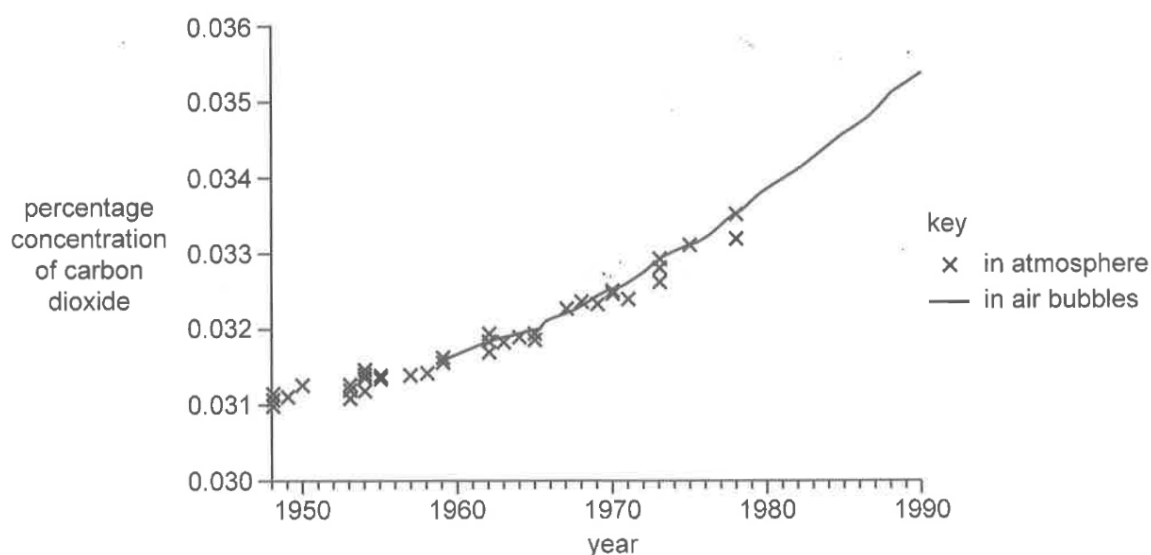


Fig. 11.1

With reference to the information provided, suggest and explain how climate change scientists can estimate atmospheric carbon dioxide concentrations 10 000 years ago.

..... [2]

Suggest

1. Scientists can analyse the concentration of carbon dioxide in air bubbles trapped deeper in the ice sheets 10 000 years ago (as an estimate of the atmospheric carbon dioxide concentrations then) ;

Explain

2. The graph in Fig. 11.1 shows that the concentration of carbon dioxide measured in air bubbles trapped in ice is a good/accurate match with the direct measurements of atmospheric carbon dioxide at the time when the air bubbles were formed ;

Idea that air bubbles could be regarded as samples of the atmosphere from the time at which they formed

- (b) Volcanic eruptions, which eject large volumes of gases such as water vapour and carbon dioxide to heights of 16 to 32 kilometres above the Earth's surface, have had an effect on the climate of the Earth.

Toba is a volcano in Sumatra that erupted approximately 12 000 years ago. Fig. 11.2 shows the location of Sumatra.

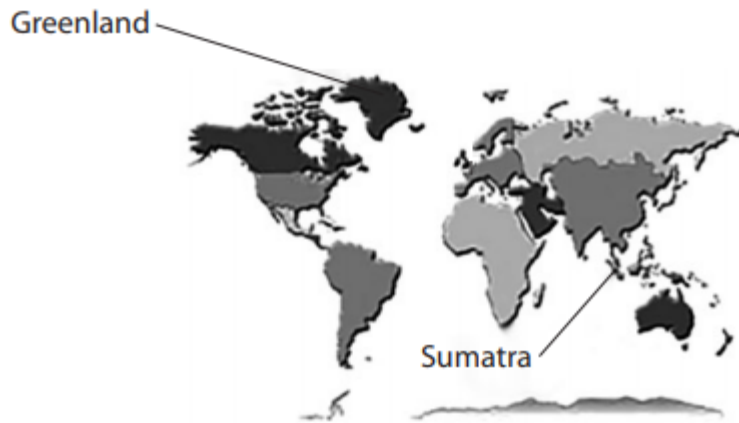


Fig. 11.2

Fig. 11.3 shows the changes to sea levels, compared to the present day, in the oceans around Greenland.

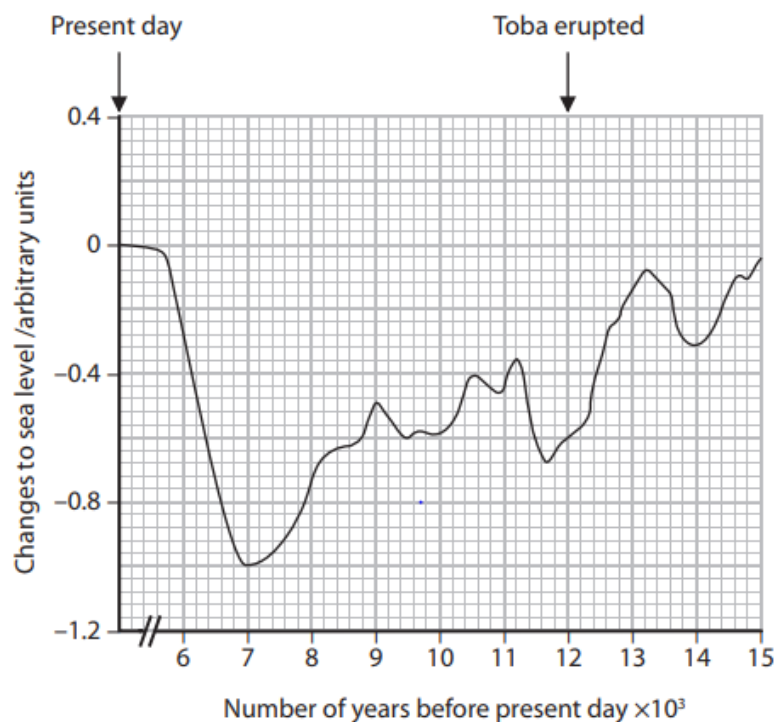


Fig. 11.3

It is claimed that the volcanic eruption of Toba caused a change in world climate.

(i) Describe the evidence in Fig. 11.3 that supports this claim.

..... [1]

1. The volcanic eruption of Toba was followed by an increase in sea level 300 years later/11 700 years ago, from -0.68 arbitrary units to ____ (e.g. -0.36) arbitrary units (accept any relevant data point) ;
OR
2. The volcanic eruption of Toba was followed by a decrease in sea level (over the next 300 years), from -0.6 arbitrary units to -0.68 arbitrary units ;

(ii) Suggest why this claim may not be true.

..... [2]

1. Prior to the eruption of Toba, fluctuations in sea level / changes in climate were already taking place (e.g. 14 000 years ago) ;
2. Changes in sea level / Climate change might be due to other factors, e.g. anthropogenic activities, rather than the eruption of Toba ;
3. The changes in sea level in Fig. 11.3 are only for one area/around Greenland but might not be applicable to the world ;

[Total: 5]